

Nutritional Value, Antinutrients and Glycemic Index of Finger Millet

Progress Report Presentation

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Food and nutrition security

- Major staples face climate pressure
- Kenya faces undernutrition and diabetes risk
- Resilient nutrient-dense crops are needed

Why finger millet

- Climate-resilient and low-input crop
- Good source of fibre, minerals and starch
- Contains bioactive compounds with health value

Research gap

- Limited Kenya-specific variety data
- Processing effects are not fully characterized
- Local GI data for finger millet foods remain limited

Main Objective

Evaluate varietal differences in nutritional composition, antinutrients and bioactive compounds in Kenyan finger millet, and determine how processing methods influence these traits and glycemic index outcomes.

Presentation note: keep this slide verbal and avoid reading full text.

Study Workflow and Current Position

Phase 1

16 varieties screened
Proximate
Minerals
Antinutrients
Bioactives

Phase 2

2 contrasting varieties
Soaking
Fermentation
Extrusion

Phase 3

Finger millet porridge
In vivo GI
Blood glucose response

Progress logic for oral presentation

- Current report should emphasize Phase 1 findings because those have measurable results.
- Processing and GI should be presented as the next analytical stage, not overloaded into current results.
- Use Phase 1 results to justify selection of candidate varieties for the next phase.

Progress to Date

Proposal refined

Done

Sample collection

Done

Proximate analysis

Done

Phytochemicals

Done

Minerals/fat

Pending

Processing trials

Next

GI testing

Next

Thesis outputs

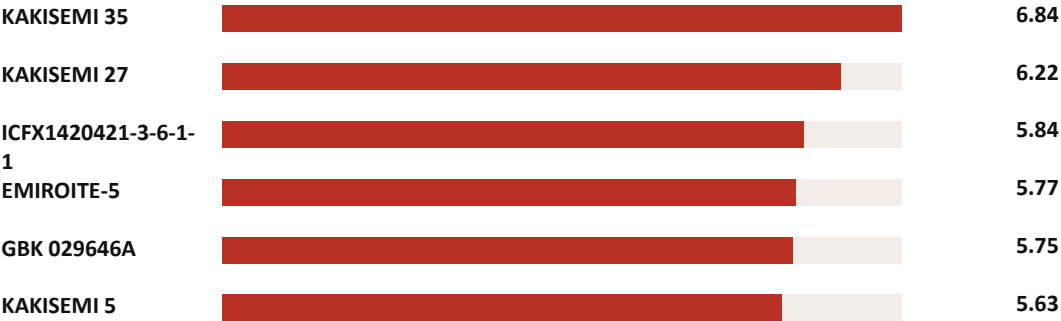
Ongoing

Progress discussion points

- Current data are strong enough to report clear varietal differences in proximate and phytochemical profile.
- The next technical step is to use these results to justify which varieties move into processing studies.
- Keep the progress report focused on achieved results plus the immediate analytical next steps.

Results Overview: Proximate Composition

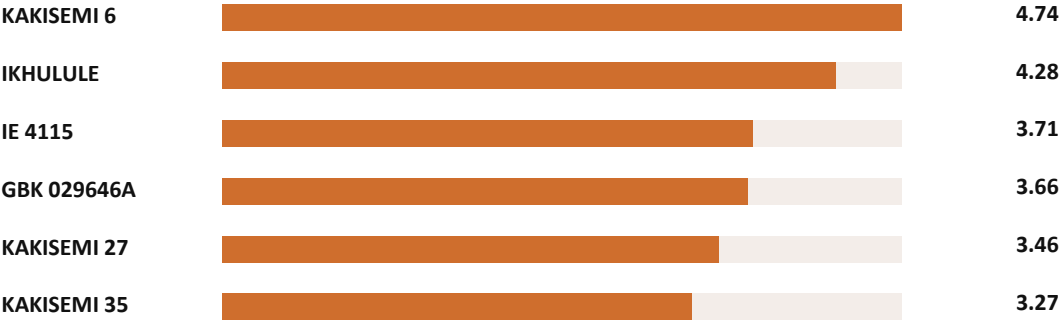
Highest protein varieties (%)



Highest crude fibre varieties (%)



Highest ash varieties (%)



Highest moisture varieties (%)

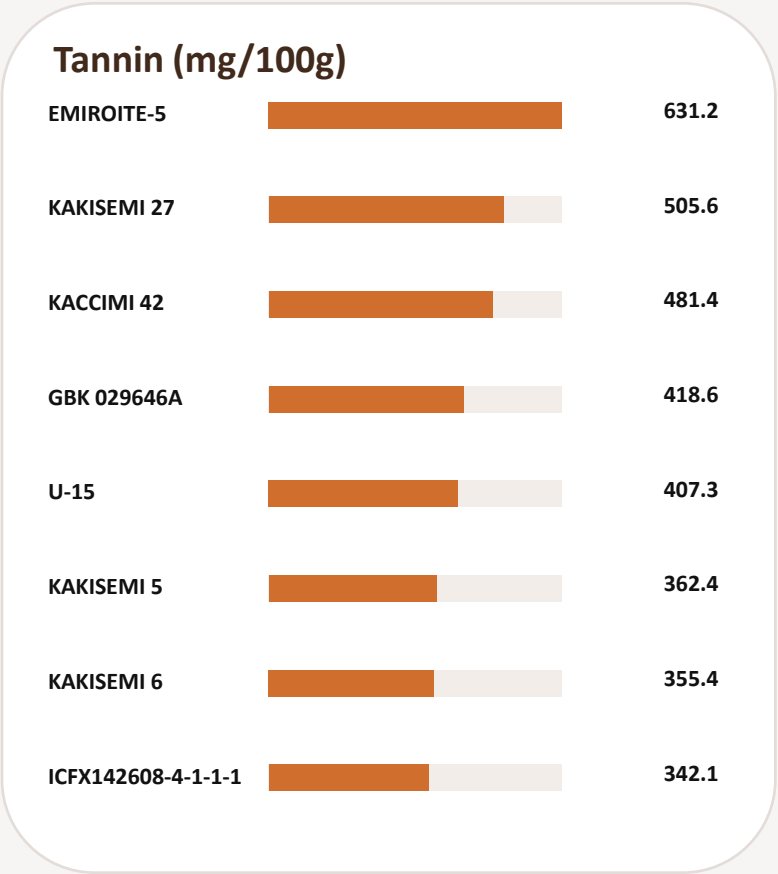
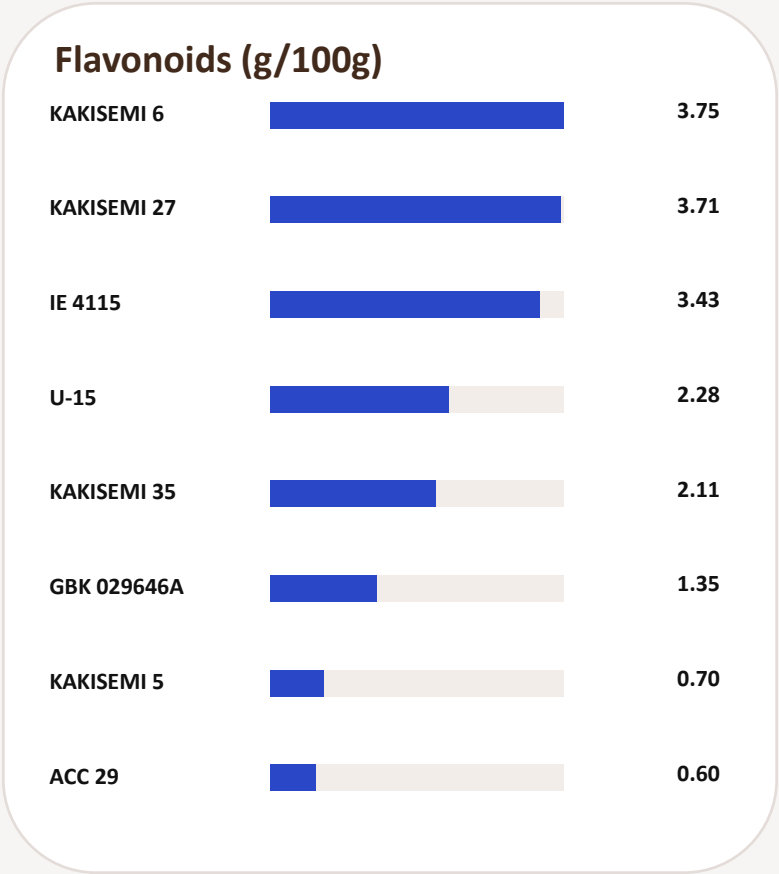
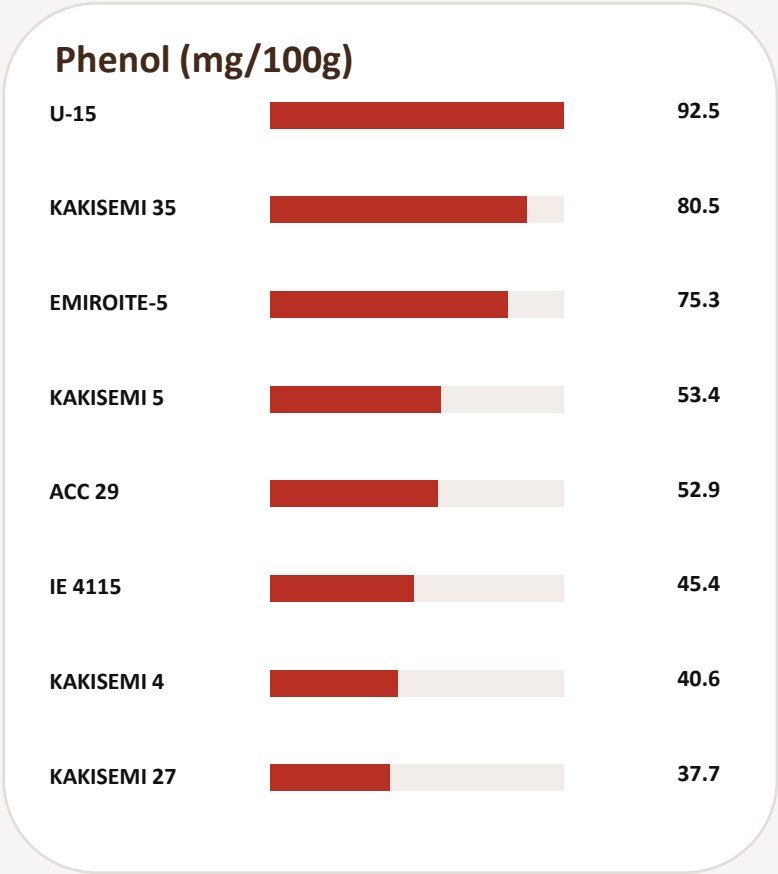


Consolidated Proximate Results Table

16 varieties

Variety	Moisture (%)	Ash (%)	Crude fibre (%)	Protein (%)
IE 4115	10.87	3.71	5.37	5.12
ACC 29	11.55	3.08	3.50	4.40
EMIROITE-5	11.41	2.51	4.33	5.77
GBK 029646A	11.08	3.66	4.68	5.75
ICFX1420421-3-6-1-1	10.60	2.89	4.89	5.84
ICFX142591-2-2-1-1	11.60	2.94	4.85	3.90
ICFX142608-4-1-1-1	11.01	2.90	5.82	5.37
IKHULULE	11.60	4.28	6.64	4.27
KACCIMI 42	11.49	1.15	5.40	4.39
KAKISEMI 15	10.98	2.79	5.76	4.99
KAKISEMI 27	10.73	3.46	3.74	6.22
KAKISEMI 35	10.61	3.27	5.48	6.84
KAKISEMI 4	11.52	3.21	4.73	5.40
KAKISEMI 5	11.22	2.82	5.79	5.63
KAKISEMI 6	11.10	4.74	3.95	3.75
U-15	10.91	2.69	4.27	4.37

Results Overview: Phytochemicals and Tannins



- U-15 and Kakisemi 35 stand out on total phenol, while Kakisemi 6 and Kakisemi 27 stand out on flavonoids.
- Emiroite-5 shows the strongest tannin signal, indicating strong phytochemical activity but also higher antinutrient burden.

Consolidated Phytochemical Results Table

16 varieties

Variety	Phenol (mg/100g)	Flavonoids (g/100g)	Tannin (mg/100g)
IE 4115	45.381	3.427	325.218
ACC 29	52.901	0.596	189.385
EMIROITE-5	75.263	0.321	631.155
GBK 029646A	36.742	1.347	418.598
ICFX1420421-3-6-1-1	33.351	0.187	302.850
ICFX142591-2-2-1-1	25.925	0.436	229.931
ICFX142608-4-1-1-1	23.466	0.351	342.078
IKHULULE	12.431	0.572	111.990
KACCIMI 42	23.475	0.333	481.383
KAKISEMI 15	29.887	0.255	171.195
KAKISEMI 27	37.712	3.707	505.561
KAKISEMI 35	80.531	2.112	333.261
KAKISEMI 4	40.610	0.160	330.111
KAKISEMI 5	53.428	0.695	362.426
KAKISEMI 6	32.256	3.750	355.384
U-15	92.511	2.278	407.323

Key Discussion Points

What the current data show

- Finger millet varieties differ meaningfully in both nutritional and phytochemical composition.
- Moisture and ash show relatively tighter clustering, while fibre, protein and phytochemical traits differentiate varieties more strongly.
- The data support variety screening as a necessary first step before processing and GI work.
- High bioactive potential must be interpreted together with tannin burden, not in isolation.

Best discussion points for oral delivery

- U-15 appears promising due to very high total phenol with acceptable protein performance.
- Kakisemi 27 and Kakisemi 6 stand out for flavonoids and may be good contrasting candidates.
- Ikhulule is notable for fibre and low phenol/tannin, which may make it useful as a contrasting low-antinutrient comparator.
- These trends justify moving to processing trials on selected contrasting varieties rather than all sixteen.

Next Steps and Reporting Close

Immediate next analyses

- Complete mineral and fat analysis
- Finalize selection of 2 contrasting varieties
- Prepare processing experiment materials

Phase 2 direction

- Apply soaking, fermentation and extrusion
- Track changes in tannins, phytates, phenols and flavonoids
- Identify the method that best improves nutritional quality

Phase 3 close

- Run in vivo glycemic index on optimized porridge
- Compare blood glucose response with reference food
- Use results to support low-GI finger millet recommendations

Close with one sentence: current results confirm strong varietal differences and provide a defensible basis for the next processing and GI stages.